

CASE-RA-D2T-A

Clinician & Researcher Brief: A Data-Grounded Path Toward Remission



Prepared: 2026-07-22

Case: fully de-identified synthetic research case

Evidence reviewed: 100 distinct PubMed documents, 87 arXiv papers, current EULAR/ACR guidance, regulatory safety information, 14 GEO series, and two locally evaluated anti-TNF response cohorts.

This document is a research briefing, not medical advice or a treatment prescription. It is designed to support a discussion with a licensed rheumatologist and RA researcher. No medication should be started, stopped, restarted, or changed from this report.

Top 10 actions for the best chance of remission

Confidence means confidence that the action improves the quality of the remission pathway for this case—not the probability that the action itself will cause remission.

Rank	Action for clinician/research discussion	Confidence	Why it ranks here
1	Confirm whether active synovial inflammation is present now	Very high	Treatment escalation only helps if inflammation is the true target; joint examination, a validated activity score, CRP/ESR, and selective imaging can separate inflammatory RA from damage, fibromyalgia, OA, neuropathic pain, or other mimics.
2	Re-establish continuous specialist-led DMARD care	Very high	The documented approximately six-month DMARD gap is a major modifiable risk. A prompt rheumatology review can restore disease-modifying care safely without unsupervised restarting.
3	Reconstruct the complete DMARD and biologic timeline	Very high	Only certolizumab is documented. Dose, duration, adherence, response, adverse effects, and discontinuation reasons determine whether this is true multidrug resistance, inadequate exposure, intolerance, or an access failure.
4	Define remission and run a formal treat-to-target schedule	Very high	Record SDAI/CDAI/DAS28 at baseline; look for substantial improvement by about three months and remission or agreed target by about six months; reassess when milestones are missed.
5	Identify the exact IBD phenotype and coordinate one cross-specialty plan	High	Crohn disease, ulcerative colitis, and other bowel phenotypes have different treatment evidence. Rheumatology and gastroenterology should jointly avoid competing or ineffective regimens.
6	Complete the mechanism-specific safety gate	High	Cardiac workup is unresolved and infection, thrombosis, malignancy, lung, pregnancy, and vaccination variables are incomplete. These can reverse the risk-benefit ranking, especially for JAK inhibitors.
7	If certolizumab was a genuine inflammatory TNFi failure, discuss switching mechanism	High	Comparative evidence after a first TNFi failure favours a non-TNF mechanism over automatic TNFi cycling in the studied populations, but this is conditional on a real adequate failure.

Rank	Action for clinician/research discussion	Confidence	Why it ranks here
8	Diagnose and treat non-inflammatory symptom amplifiers in parallel	High	Sleep apnea, depression, central sensitization, deconditioning, structural damage, and neurologic disease can sustain pain and fatigue even when synovitis improves. More immunosuppression alone will not solve them.
9	Add individualized exercise, physical therapy, and rehabilitation	High for function; moderate for remission	Exercise has the strongest integrative-guideline recommendation and improves fitness, function, pain, and fatigue. Cardiac and neurologic clearance should shape intensity in this case.
10	Use diet, smoking cessation if applicable, sleep care, and healthy body composition as adjuncts	Moderate	A Mediterranean-style pattern and other healthy behaviours may improve symptoms and cardiometabolic health. No diet, supplement, or lifestyle program has been shown to cure established RA.

One-page case interpretation

This case describes an adult woman with approximately ten years of severe, incompletely controlled rheumatoid arthritis in a multi-system autoimmune phenotype. IBD is documented, but its subtype and activity are not. Prior certolizumab pegol exposure is documented, but the reason it stopped and whether it worked are unknown. Current DMARD therapy appears to have been absent for approximately six months. Cardiac evaluation is unresolved, neurologic/bulbar evaluation is active, and a sleep study was never completed.

The case should be described as a difficult-to-treat RA candidate, not yet as confirmed EULAR difficult-to-treat RA. The formal definition requires failure of at least two biologic or targeted-synthetic DMARDs with different mechanisms after conventional DMARD failure, continuing active/progressive disease features, and management that remains problematic. The current record does not establish that history.

The strongest conclusion is not “use drug X.” It is that the current uncertainty is clinically actionable. Confirming inflammation, reconstructing treatment exposure, identifying the IBD phenotype, and completing the safety screen can eliminate inappropriate options and reveal whether a mechanism switch is justified.

What remission should mean

Remission should be measured, not inferred from a vague improvement. The 2022 ACR/EULAR revision supports Boolean 2.0 remission and SDAI/CDAI index definitions. A treating team can choose the measure, record it before therapy, and repeat it at planned checkpoints.

Current EULAR recommendations retain the treat-to-target milestones: more than 50% disease-activity improvement at approximately three months and the agreed remission/low-disease-activity target by approximately six months. Missing a milestone should trigger a structured reassessment of diagnosis, adherence, exposure, intolerance, safety, and mechanism.

For longstanding disease, pain and disability may persist after inflammation improves. That does not make remission meaningless; it means inflammatory control and rehabilitation/pain mechanisms must be measured separately.

Treatment hypotheses to discuss—not prescriptions

Hypothesis 1: the apparent refractory state is partly non-inflammatory

Confidence: established diagnostic principle. Fits the case because: disease-activity scores, CRP/ESR, imaging, and fibromyalgia assessment are all missing; sleep and neurologic workups remain incomplete. Would support it: high symptom burden with little swollen-joint or imaging inflammation. Would weaken it: persistent objective synovitis that tracks symptoms. Research/clinical implication: establish inflammation before escalating immunosuppression, while treating pain, sleep, mood, damage, and function in parallel.

Hypothesis 2: treatment interruption and access are major drivers

Confidence: high. Fits the case because: an approximately six-month DMARD gap is documented. Would support it: prior disease control followed by worsening after interruption, incomplete access, or unreliable monitoring. Would weaken it: verified continuous adequate treatment with persistent objective inflammation. Research/clinical implication: continuity may be a more powerful intervention than another biomarker search.

Hypothesis 3: certolizumab had secondary pharmacokinetic failure

Confidence: emerging and case-dependent. Fits the case because: the failure mode is unknown. Antidrug antibodies and low exposure are associated with lower response in prospective biologic cohorts. Would support it: an initial response followed by loss, low drug level, antidrug antibodies, or interrupted dosing. Would weaken it: no response despite adequate exposure or a prohibitive adverse event. Research/clinical implication: retrieve old records and existing results; consider selective—not routine—drug-level/antibody testing only if it will change a decision.

Hypothesis 4: true TNF-mechanism failure warrants a different mechanism

Confidence: high at the strategy level, not at the individual-drug level. Fits the case if: certolizumab was an adequate TNFi trial and objective inflammatory activity persisted. Would support it: verified primary non-response with adherence and dose confirmed. Would weaken it: non-inflammatory symptoms, inadequate exposure, or non-medical discontinuation after benefit. Research/clinical implication: a randomized 300-person trial found 69% versus 52% good/moderate EULAR response at 24 weeks for a non-TNF biologic versus a second TNFi. This supports a mechanism-change discussion, not a named prescription.

Hypothesis 5: RA and IBD can be managed through a coordinated shared mechanism

Confidence: emerging and phenotype-dependent. Fits the case because: both conditions are documented and some advanced therapies have overlapping indications. Would support it: confirmed active RA and an IBD phenotype for which the same mechanism is evidence-based and safe. Would weaken it: inactive or misclassified IBD, prior class failure, safety conflict, or a mechanism effective for only one disease. Research/clinical implication: cross-indication logic is valuable only after gastroenterology defines the bowel disease and prior exposure.

Hypothesis 6: synovial molecular endotyping can improve selection

Confidence: emerging/research-supported. Fits the case if: objectively active synovitis persists after multiple genuine failures and an expert biopsy program is available. Evidence: PEAC identified lympho-myeloid, diffuse-myeloid, and pauci-immune/fibroblastic synovial pathotypes. The 2025 STRAP analysis reported therapy-specific RNA models with external validation for tocilizumab and rituximab and stronger performance after conversion to a synovial panel. Limitation: this is not yet a universally available, prospectively proven prescribing test. Research implication: this case is more suited to a synovial precision-research pathway than to another small blood-only classifier.

Hypothesis 7: immune-reset or cell therapy could become a future option

Confidence: investigational. Fits only if: formal multidrug-resistant inflammatory disease is proven and a legitimate trial accepts the patient. Research implication: track trials, but do not treat early autoimmune cell-therapy reports as evidence of established RA remission for this case.

Safety logic that changes the ranking

The unresolved cardiac workup prevents this report from favouring a JAK inhibitor. In ORAL Surveillance, cardiovascular-risk-enriched RA patients receiving tofacitinib had more major cardiovascular events and cancers than TNF-inhibitor recipients; non-inferiority was not established. FDA boxed warnings also cover serious heart-related events, cancer, blood clots, and death for specified JAK inhibitors.

This is not a declaration that every JAK inhibitor is inappropriate. It is a declaration that this case lacks the information needed to rank that class safely. Smoking, cardiovascular history, thrombosis, malignancy, infection, vaccination, lung disease, blood counts, kidney/liver status, pregnancy plans if relevant, and medication interactions must be integrated.

The IBD diagnosis also changes the safety and efficacy matrix. A therapy effective in RA may be ineffective or undesirable for a particular bowel phenotype; an IBD therapy may not control RA. Cross-specialty review should occur before—not after—mechanism selection.

Diet and lifestyle: direct answer about cure

No reviewed document demonstrated that diet or lifestyle can cure established rheumatoid arthritis. ACR positions integrative interventions alongside DMARDs. EULAR's diet evidence synthesis found moderate evidence of small effects for a few dietary exposures and no high-level evidence of clinically meaningful disease modification.

Diet evidence

Approach	What the research showed	Meaning for this case
Mediterranean-style pattern	Conditional guideline support; small studies suggest possible symptom and cardiometabolic benefit	Reasonable default if compatible with the IBD phenotype and nutritional status; not a cure
Anti-inflammatory diets	Seven-trial meta-analysis found pain about 9 mm lower on a 100-mm scale, but all studies were high risk of bias and certainty was very low	Possible modest adjunct; no proof of remission or structural protection
ADIRA portfolio diet	The prespecified adjusted DAS28-ESR comparison was not significant; a completer/unadjusted analysis favoured intervention	Interesting signal, not definitive efficacy
Omega-3	Reviews suggest small improvements in some outcomes	Food-first discussion is reasonable; supplement dose, bleeding, GI tolerance, quality, and interactions require review
Vegan, vegetarian, or elimination diets	Heterogeneous, small, individualized responses	No proven remission strategy; restrictive diets may be risky with IBD or undernutrition
Fasting	Transient subjective improvements in some older studies	Not durable disease modification and not a safe unsupervised strategy
Ketogenic diet	Mostly mechanistic/scoping evidence	No convincing RA remission evidence; potential lipid, GI, and nutritional concerns
Gluten-free diet	No RA cure evidence without a separate gluten-related diagnosis	Use only for confirmed celiac disease, allergy, or another supported indication
Probiotics/microbiome products	Biologically plausible but heterogeneous early evidence	Investigational; commercial testing cannot select RA treatment
Herbal/supplement stacks	Variable formulations, purity, interactions, and publication bias	Do not replace DMARDs; disclose every product to clinicians and pharmacists

Because the IBD phenotype is unknown, a generic high-fiber, elimination, fasting, ketogenic, or probiotic plan could be inappropriate. A gastroenterology-aware registered dietitian should assess activity, strictures or surgery, deficiencies, weight trajectory, anemia, protein needs, and tolerated foods.

Exercise, rehabilitation, and function

Consistent exercise is the only strong recommendation in the ACR integrative RA guideline. EULAR recommends physical activity as standard care with individualized dose, adaptation, behaviour support, and less sedentary time. Systematic reviews show benefits in aerobic capacity, strength, function, pain, and fatigue; RA disease activity is not consistently changed.

For this case, unresolved cardiac and neurologic/bulbar workups should guide clearance and intensity. Physical therapy can adapt aerobic, resistance, aquatic, range-of-motion, balance, and hand programs. Exercise improves resilience and function; it does not cure RA.

Smoking, body composition, sleep, and mental health

- If the patient smokes, cessation is a high-value action because smoking is associated with worse cardiovascular outcomes, higher RA activity/severity, and poorer treatment response.
- Weight should not be assumed. Higher adiposity can worsen outcomes, while RA and IBD can cause sarcopenia or undernutrition. The goal is healthy body composition, adequate protein, and safe muscle-preserving activity.
- Completing the sleep evaluation can clarify fatigue, pain amplification, mood, and cardiovascular risk. Sleep improvement is supportive, not disease-eradicating.
- Cognitive-behavioural approaches, pacing, joint protection, occupational therapy, and self-management can improve participation, pain, fatigue, and adherence without implying that stress caused the autoimmune disease.

What the local experiments taught us

Experiment 1: GSE15258 anti-TNF blood-expression baseline

Question: Can baseline blood expression distinguish anti-TNF response? Data: 46 publication-matched patients: 24 responders and 22 non-responders; **MEDIUM** outcomes excluded. Models: regularized logistic regression, random forest, CPU XGBoost.

Model	Held-in ROC-AUC	Sealed ROC-AUC
Logistic regression	0.526	0.600
Random forest	0.539	0.560
XGBoost	0.551	0.640

Lesson: the signal was weak and intervals overlapped chance. XGBoost had the best point estimates, but the ten-person sealed set was too small for treatment conclusions.

Experiment 2: three XGBoost challengers

Changes tested: reduce selected probes from 200 to 40, reduce depth from 3 to 2, and increase L2 regularization from 2 to 10. Result: every candidate improved one partition while worsening the other. Lesson: apparent gains in a small cohort were validation instability, not robust improvement. All challengers were rejected under the fixed non-regression gate.

Experiment 3: GSE129705 independent-cohort evaluation

Question: Does the signal generalize to a separate baseline whole-blood RNA-seq cohort? Data: Cohort 1 development population, 34 patients; Cohort 2 sealed external population, 29 patients; EULAR **Good** versus **None**.

Model	Held-in ROC-AUC	External ROC-AUC	External 95% interval
Logistic regression	0.708	0.662	0.443-0.851
Random forest	0.691	0.606	0.374-0.818
XGBoost	0.667	0.636	0.406-0.841

Lesson: regularized logistic regression ranked highest, but the external interval included chance. This is exploratory population signal, not a patient-level selector.

Experiment 4: three external-cohort logistic challengers

Changes tested: 20 selected genes, 100 selected genes, and stronger L2 regularization. Result: all regressed on both or one fixed population. Lesson: generic hyperparameter tuning was exhausted. Continuing to choose against the external cohort would create leakage.

Experiment 5: arXiv treatment-research loop

Scope: all 87 unique papers returned by the current broad RA query; accessible PDFs read locally. Result: 33 treatment-screen candidates, 9 papers with a named data source or identifier, and 14 unique GEO series checked. No new compatible public baseline treatment-response cohort was found. Lesson: the limiting factor is not a shortage of models. It is the shortage of public, well-labeled, independently validated response cohorts.

Experiment 6: current guidance, trials, and regulator review

Sources: current EULAR RA guidance, EULAR D2T points, ACR context, switch-versus-cycle trial evidence, network meta-analysis, SELECT-SWITCH, FDA JAK warnings, and trial registries. Lesson: after a genuine first TNFi failure with active RA, changing mechanism has stronger comparative support than automatic cycling in studied populations. The evidence remains conditional on failure mode and safety.

Experiment 7: corrected 100-document PubMed loop

Design: 25 remission/treat-to-target, 25 treatment-sequence/precision, 20 IBD/safety, 15 diet/nutrition, and 15 lifestyle/function documents. Sources were unique by PMID and normalized title. Read: 150,994 abstract characters and 40 locally cached open full texts. Evidence mix: 2 guidelines, 12 systematic reviews/meta-analyses, 6 randomized trials, 3 observational/registry studies, 18 other peer-reviewed studies, and 59 reviews. Quality lesson: the first newest-first retrieval was rejected because it admitted incidental and preclinical papers. Requiring RA-indexed human evidence, title-level topic relevance, readable abstracts, and relevance ranking produced the final evidence set. Clinical lesson: no drug or lifestyle cure emerged. The top hypotheses stabilized around inflammation confirmation, treatment-history reconstruction, continuity, coordinated RA-IBD selection, individualized safety, and treat-to-target monitoring.

What the research documents collectively taught us

Established findings

- Confirm inflammation and diagnosis before escalating D2T RA therapy.
- Formal D2T status cannot be assigned from one documented biologic exposure.
- Treat-to-target monitoring and rapid strategy adjustment matter more than unstructured drug cycling.
- After a genuine first TNFi failure, a different mechanism deserves serious consideration.
- Exercise improves function and health and should accompany pharmacologic care.
- Smoking, cardiovascular risk, infection risk, sleep, mental health, and treatment access affect outcomes.
- Diet and lifestyle do not replace DMARDs and have not cured established RA.

Emerging findings

- Synovial molecular signatures may predict differential response better than generic blood signatures.
- Antidrug antibodies and low exposure can explain some secondary biologic failures.
- Coordinating one mechanism across RA and a defined IBD phenotype may reduce therapeutic conflict.
- The microbiome and dietary patterns may modify symptoms or inflammatory pathways, but actionable patient-level tests are not validated.

Investigational findings

- CAR-T and other immune-reset strategies are trial-only possibilities.
- Novel nanoparticles, cell products, metabolic targets, and repurposed compounds frequently appear in preclinical literature but do not establish human remission.
- Machine-learning treatment selection remains limited by small cohorts, inconsistent biology, and lack of external prospective validation.

Research opportunities for an RA investigator

- 1** Prospective deep phenotyping: create a cohort with complete exposure, adherence, response, discontinuation, safety, IBD, imaging, and patient-reported data.
- 2** Synovial-to-blood translation: prospectively test whether validated synovial pathotypes can be approximated from accessible blood/cell-composition features without losing treatment specificity.
- 3** RA-IBD joint cohort: study patients with both diseases using outcomes from both specialties rather than treating the overlap as an exclusion.
- 4** Failure-mode labeling: distinguish never-exposed, pharmacokinetic failure, primary non-response, secondary loss, intolerance, and access interruption in every dataset.
- 5** Pragmatic mechanism-sequence trials: compare switch strategies after two or more mechanisms while stratifying by serology, synovial pathotype, comorbidity, and reason for failure.
- 6** Diet as adjunct, not replacement: test Mediterranean-style and IBD-compatible dietary support with adequate power, objective inflammatory endpoints, and medication stability.
- 7** Continuity interventions: measure whether navigation, rapid access, pharmacy support, and nurse-led monitoring reduce gaps and improve treat-to-target success.

Clinician/researcher discussion checklist

- Is inflammatory activity objectively present today?
- Does the patient meet the formal D2T definition?
- What was the exact outcome and failure mode of each prior DMARD?
- Was certolizumab stopped for biology, safety, intolerance, adherence, or access?
- What is the IBD diagnosis, activity, and current treatment?
- Which mechanism can safely address the whole phenotype?
- What cardiac, thrombotic, infection, malignancy, lung, neurologic, pregnancy, and vaccination variables change the ranking?
- What measure defines remission, and who will check it at three and six months?
- Which residual symptoms need rehabilitation, sleep, mental-health, pain, or neurologic care rather than more immunosuppression?
- Is a synovial-biopsy precision program or appropriate clinical trial available if genuine multidrug resistance is confirmed?

Key references

- [EULAR 2025 RA management recommendations](#), published 2026: DOI 10.1016/j.ard.2026.01.023; PMID 41826212.
- [EULAR D2T RA definition](#): PMID 33004335.
- [EULAR D2T management points](#): PMID 34407926.
- [ACR/EULAR remission criteria revision](#): PMID 36274193.
- [Non-TNF biologic versus second TNFi randomized strategy trial](#): PMID 27654603.
- [Switch-versus-cycle Bayesian network meta-analysis](#): PMID 33854570.
- [ORAL Surveillance cardiovascular/cancer safety trial](#): PMID 35081280.
- [ACR integrative RA guideline](#): PMID 37227116.
- [EULAR lifestyle recommendations](#): PMID 35260387.
- [EULAR 2025 physical-activity update](#): PMID 42036268.
- [Diet effects systematic review informing EULAR](#): PMID 35654458.
- [Anti-inflammatory diet pain meta-analysis](#): PMID 34959772.
- [STRAP deep synovial treatment-response analysis](#): PMID 40603860.
- [Antidrug antibody prospective cohort](#): PMID 37436748.

Standing disclaimer

De-identified research aid only. Not medical advice, diagnosis, or treatment. All hypotheses, tests, medication choices, diet changes, supplements, and exercise plans require review and decision by the licensed care team.

Appendix A

Complete 100-document PubMed review catalogue

All entries are distinct by PMID and normalized title. The catalogue records what each document contributed to the review; it does not convert every paper into a treatment recommendation.

001 Distinct synovial tissue macrophage subsets regulate inflammation and remission in rheumatoid arthritis.

PMID 32601335 • 2020 • other peer reviewed research • remission and treat to target

Peer-reviewed evidence informing remission definitions, D2T framing, or treat-to-target implementation. Routed as other peer reviewed research; applicability was limited to the population and design reported.

002 EULAR definition of difficult-to-treat rheumatoid arthritis.

PMID 33004335 • 2021 • other peer reviewed research • remission and treat to target

Peer-reviewed evidence informing remission definitions, D2T framing, or treat-to-target implementation. Routed as other peer reviewed research; applicability was limited to the population and design reported.

003 A Review of Difficult-to-Treat Rheumatoid Arthritis: Definition, Clinical Presentation, and Management.

PMID 37776482 • 2023 • review • remission and treat to target

Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.

004 Synovial tissue macrophages in joint homeostasis, rheumatoid arthritis and disease remission.

PMID 35672464 • 2022 • review • remission and treat to target

Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.

005 EULAR points to consider for the management of difficult-to-treat rheumatoid arthritis.

PMID 34407926 • 2022 • guideline • remission and treat to target

Formal guidance informing remission definitions, D2T framing, or treat-to-target implementation. Routed as guideline; applicability was limited to the population and design reported.

006 Difficult-to-treat rheumatoid arthritis: systematic review and meta-analysis on global prevalence.

PMID 41188120 • 2026 • systematic review or meta analysis • remission and treat to target

Evidence synthesis informing remission definitions, D2T framing, or treat-to-target implementation. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

007 Difficult-to-treat rheumatoid arthritis: what have we learned and what do we still need to learn?

PMID 39383505 • 2025 • review • remission and treat to target

Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.

008 Five-year follow-up of patients with difficult-to-treat rheumatoid arthritis.

PMID 38851883 • 2025 • other peer reviewed research • remission and treat to target

Peer-reviewed evidence informing remission definitions, D2T framing, or treat-to-target implementation. Routed as other peer reviewed research; applicability was limited to the population and design reported.

009 [Difficult-to-treat rheumatoid arthritis].

PMID 38482760 • 2024 • other peer reviewed research • remission and treat to target

Peer-reviewed evidence informing remission definitions, D2T framing, or treat-to-target implementation. Routed as other peer reviewed research; applicability was limited to the population and design reported.

010 American College of Rheumatology/EULAR Remission Criteria for Rheumatoid Arthritis: 2022 Revision.

PMID 36274193 • 2023 • other peer reviewed research • remission and treat to target

Peer-reviewed evidence informing remission definitions, D2T framing, or treat-to-target implementation. Routed as other peer reviewed research; applicability was limited to the population and design reported.

- 011 Efficacy and safety of current therapies for difficult-to-treat rheumatoid arthritis: a systematic review and network meta-analysis.**
PMID 39198829 • 2024 • systematic review or meta analysis • remission and treat to target
Evidence synthesis informing remission definitions, D2T framing, or treat-to-target implementation. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.
- 012 Multi-omics monitoring of drug response in rheumatoid arthritis in pursuit of molecular remission.**
PMID 30013029 • 2018 • other peer reviewed research • remission and treat to target
Peer-reviewed evidence informing remission definitions, D2T framing, or treat-to-target implementation. Routed as other peer reviewed research; applicability was limited to the population and design reported.
- 013 POST-COVID-19 INFLAMMATORY RHEUMATOID ARTHRITIS REMISSION.**
PMID 38807392 • 2024 • review • remission and treat to target
Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.
- 014 Treat-to-target: measures.**
PMID 23073309 • 2012 • other peer reviewed research • remission and treat to target
Peer-reviewed evidence informing remission definitions, D2T framing, or treat-to-target implementation. Routed as other peer reviewed research; applicability was limited to the population and design reported.
- 015 Remission in rheumatoid arthritis.**
PMID 20697983 • 2010 • review • remission and treat to target
Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.
- 016 Non-pharmacological treatment in difficult-to-treat rheumatoid arthritis.**
PMID 36106320 • 2022 • review • remission and treat to target
Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.
- 017 Defining remission in rheumatoid arthritis.**
PMID 22460146 • 2012 • other peer reviewed research • remission and treat to target
Peer-reviewed evidence informing remission definitions, D2T framing, or treat-to-target implementation. Routed as other peer reviewed research; applicability was limited to the population and design reported.
- 018 Difficult-to-treat rheumatoid arthritis: Current concept and unsolved problems.**
PMID 36353219 • 2022 • review • remission and treat to target
Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.
- 019 [Treat-to-target in rheumatology].**
PMID 26753210 • 2015 • review • remission and treat to target
Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.
- 020 Treat-to-target in rheumatoid arthritis - are we there yet?**
PMID 30700865 • 2019 • review • remission and treat to target
Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.
- 021 Remission in rheumatoid arthritis: is it all the same?**
PMID 26131966 • 2015 • review • remission and treat to target
Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.
- 022 Treat-to-target in systemic lupus erythematosus: advancing towards its implementation.**
PMID 35039665 • 2022 • review • remission and treat to target
Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.

023 Rheumatoid Arthritis: Defining Clinical and Ultrasound Deep Remission.

PMID 33521569 • 2020 • review • remission and treat to target

Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.

024 Clinical remission in rheumatoid arthritis and psoriatic arthritis.

PMID 29600949 • 2018 • review • remission and treat to target

Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.

025 Management of Late-Onset Rheumatoid Arthritis with Treat-to-Target Strategy.

PMID 40202700 • 2025 • review • remission and treat to target

Review used to map remission definitions, D2T framing, or treat-to-target implementation. Routed as review; applicability was limited to the population and design reported.

026 Upadacitinib: Mechanism of action, clinical, and translational science.

PMID 37984057 • 2024 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

027 Molecular mechanism of action and pharmacokinetic properties of methotrexate.

PMID 32415503 • 2020 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

028 Clinical effect and biological mechanism of exercise for rheumatoid arthritis: A mini review.

PMID 36685485 • 2022 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

029 Economic burden of rheumatoid arthritis: a systematic review of literature in biologic era.

PMID 32245893 • 2020 • systematic review or meta analysis • treatment sequence and precision

Evidence synthesis informing mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

030 Molecular Portraits of Early Rheumatoid Arthritis Identify Clinical and Treatment Response Phenotypes.

PMID 31461658 • 2019 • other peer reviewed research • treatment sequence and precision

Peer-reviewed evidence informing mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as other peer reviewed research; applicability was limited to the population and design reported.

031 A longitudinal cohort study uncovers plasma protein biomarkers predating clinical onset and treatment response of rheumatoid arthritis.

PMID 40691443 • 2025 • observational or registry • treatment sequence and precision

Population/registry evidence informing mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as observational or registry; applicability was limited to the population and design reported.

032 Precision medicine in rheumatoid arthritis.

PMID 35248489 • 2022 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

033 Matrix metalloproteinases in arthritis: towards precision medicine.

PMID 37161083 • 2023 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

034 Tapering biologic DMARDs in rheumatoid arthritis.

PMID 36274358 • 2022 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

035 Response to Biologic Drugs in Patients With Rheumatoid Arthritis and Antidrug Antibodies.

PMID 37436748 • 2023 • other peer reviewed research • treatment sequence and precision

Peer-reviewed evidence informing mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as other peer reviewed research; applicability was limited to the population and design reported.

036 Predictors of treatment response in rheumatoid arthritis.

PMID 29981377 • 2019 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

037 A Longitudinal Study Reveals Metabolomic Markers for Individuals at Risk, Disease Severity, and Treatment Response in Rheumatoid Arthritis.

PMID 40801465 • 2025 • other peer reviewed research • treatment sequence and precision

Peer-reviewed evidence informing mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as other peer reviewed research; applicability was limited to the population and design reported.

038 Precision medicine and management of rheumatoid arthritis.

PMID 32276742 • 2020 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

039 -Omic Approaches and Treatment Response in Rheumatoid Arthritis.

PMID 36015273 • 2022 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

040 Precision medicine: the precision gap in rheumatic disease.

PMID 36216923 • 2022 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

041 Biologic therapies and bone loss in rheumatoid arthritis.

PMID 27796445 • 2017 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

042 Pharmacogenomics of biological treatment in rheumatoid arthritis.

PMID 24295560 • 2014 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

043 Non-TNF biologics and their biosimilars in rheumatoid arthritis.

PMID 38766765 • 2024 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

044 Predicting treatment response to IL6R blockers in rheumatoid arthritis.

PMID 32864695 • 2020 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

045 Tailoring biologic therapy for real-world rheumatoid arthritis patients.

PMID 33147106 • 2021 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

046 Rheumatoid arthritis: Perioperative management of biologics and DMARDs.

PMID 25747348 • 2015 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

047 Rheumatoid arthritis and the biological clock.

PMID 24684672 • 2014 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

048 Deep molecular profiling of synovial biopsies in the STRAP trial identifies signatures predictive of treatment response to biologic therapies in rheumatoid arthritis.

PMID 40603860 • 2025 • randomized trial • treatment sequence and precision

Controlled trial evidence informing mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as randomized trial; applicability was limited to the population and design reported.

049 Biologic interventions for fatigue in rheumatoid arthritis.

PMID 27271314 • 2016 • systematic review or meta analysis • treatment sequence and precision

Evidence synthesis informing mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

050 Pursuing Precision Medicine in Managing Rheumatoid Arthritis.

PMID 40269471 • 2025 • review • treatment sequence and precision

Review used to map mechanism sequencing, treatment response, biomarkers, or precision medicine. Routed as review; applicability was limited to the population and design reported.

051 Cardiovascular and Cancer Risk with Tofacitinib in Rheumatoid Arthritis.

PMID 35081280 • 2022 • randomized trial • ibd overlap and safety

Controlled trial evidence informing comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as randomized trial; applicability was limited to the population and design reported.

052 Cardiovascular Manifestations in Rheumatoid Arthritis.

PMID 36729119 • 2022 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

053 Efficacy and safety of JAK inhibitors in rheumatoid arthritis: update for the practising clinician.

PMID 38216757 • 2024 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

054 Increased cardiovascular risk in rheumatoid arthritis: mechanisms and implications.

PMID 29685876 • 2018 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

055 Shared inflammatory pathways of rheumatoid arthritis and atherosclerotic cardiovascular disease.

PMID 37231248 • 2023 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

056 Safety Profile of Upadacitinib: Descriptive Analysis in Over 27,000 Patient-Years Across Rheumatoid Arthritis, Psoriatic Arthritis, Axial Spondyloarthritis, Atopic Dermatitis, and Inflammatory Bowel Disease.

PMID 40875187 • 2025 • other peer reviewed research • ibd overlap and safety

Peer-reviewed evidence informing comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as other peer reviewed research; applicability was limited to the population and design reported.

057 Cardiovascular Safety of Tocilizumab Versus Etanercept in Rheumatoid Arthritis: A Randomized Controlled Trial.

PMID 31469238 • 2020 • randomized trial • ibd overlap and safety

Controlled trial evidence informing comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as randomized trial; applicability was limited to the population and design reported.

058 Cardiovascular complications of rheumatoid arthritis.

PMID 38334476 • 2024 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

059 Cardiovascular Risk Factor and Atherosclerosis in Rheumatoid Arthritis (RA).

PMID 39831939 • 2025 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

060 Cardiovascular disease in rheumatoid arthritis.

PMID 24268008 • 2014 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

061 Rheumatoid arthritis (RA) and cardiovascular disease.

PMID 31059840 • 2019 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

062 Rheumatoid arthritis and cardiovascular disease.

PMID 24093840 • 2013 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

063 [Cardiovascular comorbidities in rheumatoid arthritis].

PMID 30656399 • 2019 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

064 Atherosclerotic cardiovascular disease prevention in rheumatoid arthritis.

PMID 32494054 • 2020 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

065 Glucocorticoid safety for treating rheumatoid arthritis.

PMID 25802019 • 2015 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

066 Cardiovascular and Metabolic Comorbidities in Rheumatoid Arthritis.

PMID 30397830 • 2018 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

067 Rheumatoid arthritis and risk of cardiovascular disease.

PMID 29583150 • 2018 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

068 Cardiovascular Risk in Rheumatoid Arthritis: Literature Review.

PMID 34164156 • 2021 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

069 Exploring candidate biomarkers for rheumatoid arthritis through cardiovascular and cardiometabolic serum proteome profiling.

PMID 38420123 • 2024 • other peer reviewed research • ibd overlap and safety

Peer-reviewed evidence informing comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as other peer reviewed research; applicability was limited to the population and design reported.

070 Cardiovascular Disease Risk in Rheumatoid Arthritis Anno 2022.

PMID 35628831 • 2022 • review • ibd overlap and safety

Review used to map comorbidity, IBD overlap, cardiovascular risk, infection, or treatment safety. Routed as review; applicability was limited to the population and design reported.

071 2022 American College of Rheumatology Guideline for Exercise, Rehabilitation, Diet, and Additional Integrative Interventions for Rheumatoid Arthritis.

PMID 37227116 • 2023 • guideline • diet and nutrition

Formal guidance informing diet, nutrition, supplements, or microbiome evidence. Routed as guideline; applicability was limited to the population and design reported.

072 Effect of Anti-Inflammatory Diets on Pain in Rheumatoid Arthritis: A Systematic Review and Meta-Analysis.

PMID 34959772 • 2021 • systematic review or meta analysis • diet and nutrition

Evidence synthesis informing diet, nutrition, supplements, or microbiome evidence. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

073 Nutrition and its role in prevention and management of rheumatoid arthritis.

PMID 37182439 • 2023 • review • diet and nutrition

Review used to map diet, nutrition, supplements, or microbiome evidence. Routed as review; applicability was limited to the population and design reported.

074 Dietary Habits and Nutrition in Rheumatoid Arthritis: Can Diet Influence Disease Development and Clinical Manifestations?

PMID 32443535 • 2020 • review • diet and nutrition

Review used to map diet, nutrition, supplements, or microbiome evidence. Routed as review; applicability was limited to the population and design reported.

075 Nutrition and Diet in Rheumatoid Arthritis.

PMID 35215538 • 2022 • other peer reviewed research • diet and nutrition

Peer-reviewed evidence informing diet, nutrition, supplements, or microbiome evidence. Routed as other peer reviewed research; applicability was limited to the population and design reported.

076 Dietary-timing-induced gut microbiota diurnal oscillations modulate inflammatory rhythms in rheumatoid arthritis.

PMID 39260371 • 2024 • other peer reviewed research • diet and nutrition

Peer-reviewed evidence informing diet, nutrition, supplements, or microbiome evidence. Routed as other peer reviewed research; applicability was limited to the population and design reported.

077 Rheumatoid arthritis and dietary interventions: systematic review of clinical trials.

PMID 32585000 • 2021 • systematic review or meta analysis • diet and nutrition

Evidence synthesis informing diet, nutrition, supplements, or microbiome evidence. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

078 Rheumatoid arthritis and the intestinal microbiome: probiotics as a potential therapy.

PMID 38510244 • 2024 • review • diet and nutrition

Review used to map diet, nutrition, supplements, or microbiome evidence. Routed as review; applicability was limited to the population and design reported.

079 Anti-inflammatory Diet In Rheumatoid Arthritis (ADIRA)-a randomized, controlled crossover trial indicating effects on disease activity.

PMID 32055820 • 2020 • randomized trial • diet and nutrition

Controlled trial evidence informing diet, nutrition, supplements, or microbiome evidence. Routed as randomized trial; applicability was limited to the population and design reported.

080 The Role of Diet in Modulating Inflammation and Oxidative Stress in Rheumatoid Arthritis, Ankylosing Spondylitis, and Psoriatic Arthritis.

PMID 40362911 • 2025 • review • diet and nutrition

Review used to map diet, nutrition, supplements, or microbiome evidence. Routed as review; applicability was limited to the population and design reported.

081 A high-fiber diet synergizes with Prevotella copri and exacerbates rheumatoid arthritis.

PMID 36323929 • 2022 • other peer reviewed research • diet and nutrition

Peer-reviewed evidence informing diet, nutrition, supplements, or microbiome evidence. Routed as other peer reviewed research; applicability was limited to the population and design reported.

082 Dietary Interventions with or without Omega-3 Supplementation for the Management of Rheumatoid Arthritis: A Systematic Review.

PMID 34684507 • 2021 • systematic review or meta analysis • diet and nutrition

Evidence synthesis informing diet, nutrition, supplements, or microbiome evidence. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

083 Nutrition and diet in rheumatoid arthritis, axial spondyloarthritis, and psoriatic arthritis: a systematic review.

PMID 41030267 • 2025 • systematic review or meta analysis • diet and nutrition

Evidence synthesis informing diet, nutrition, supplements, or microbiome evidence. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

084 Efficacy and safety of dietary polyphenols in rheumatoid arthritis: A systematic review and meta-analysis of 47 randomized controlled trials.

PMID 37033930 • 2023 • systematic review or meta analysis • diet and nutrition

Evidence synthesis informing diet, nutrition, supplements, or microbiome evidence. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

085 Effects of omega-3 supplementation on lipid metabolism, inflammation, and disease activity in rheumatoid arthritis: a meta-analysis of randomized controlled trials.

PMID 38922552 • 2024 • systematic review or meta analysis • diet and nutrition

Evidence synthesis informing diet, nutrition, supplements, or microbiome evidence. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

086 Physical activity, exercise and rheumatoid arthritis: Effectiveness, mechanisms and implementation.

PMID 31203925 • 2018 • review • lifestyle and function

Review used to map exercise, function, smoking, sleep, weight, or stress evidence. Routed as review; applicability was limited to the population and design reported.

087 The effect of physical exercise on rheumatoid arthritis: An overview of systematic reviews and meta-analysis.

PMID 33176012 • 2021 • systematic review or meta analysis • lifestyle and function

Evidence synthesis informing exercise, function, smoking, sleep, weight, or stress evidence. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

088 Exercise for rheumatoid arthritis of the hand.

PMID 30063798 • 2018 • systematic review or meta analysis • lifestyle and function

Evidence synthesis informing exercise, function, smoking, sleep, weight, or stress evidence. Routed as systematic review or meta analysis; applicability was limited to the population and design reported.

089 Smoking and rheumatoid arthritis.

PMID 25479074 • 2014 • review • lifestyle and function

Review used to map exercise, function, smoking, sleep, weight, or stress evidence. Routed as review; applicability was limited to the population and design reported.

090 Effects of Aerobic and Resistance Exercise in Older Adults With Rheumatoid Arthritis: A Randomized Controlled Trial.

PMID 29696812 • 2019 • randomized trial • lifestyle and function

Controlled trial evidence informing exercise, function, smoking, sleep, weight, or stress evidence. Routed as randomized trial; applicability was limited to the population and design reported.

091 Mediterranean Diet and Physical Activity Nudges versus Usual Care in Women with Rheumatoid Arthritis: Results from the MADEIRA Randomized Controlled Trial.

PMID 36771382 • 2023 • randomized trial • lifestyle and function

Controlled trial evidence informing exercise, function, smoking, sleep, weight, or stress evidence. Routed as randomized trial; applicability was limited to the population and design reported.

092 Rheumatoid Arthritis and Oxidative Stress.

PMID 36227658 • 2022 • other peer reviewed research • lifestyle and function

Peer-reviewed evidence informing exercise, function, smoking, sleep, weight, or stress evidence. Routed as other peer reviewed research; applicability was limited to the population and design reported.

093 [Stress and rheumatoid arthritis].

PMID 20652573 • 2010 • review • lifestyle and function

Review used to map exercise, function, smoking, sleep, weight, or stress evidence. Routed as review; applicability was limited to the population and design reported.

094 Physical activity of older patients with rheumatoid arthritis.

PMID 40560403 • 2025 • review • lifestyle and function

Review used to map exercise, function, smoking, sleep, weight, or stress evidence. Routed as review; applicability was limited to the population and design reported.

095 Exercise and fatigue in rheumatoid arthritis.

PMID 24575509 • 2014 • review • lifestyle and function

Review used to map exercise, function, smoking, sleep, weight, or stress evidence. Routed as review; applicability was limited to the population and design reported.

096 Rheumatoid arthritis and sleep quality.

PMID 28743355 • 2017 • observational or registry • lifestyle and function

Population/registry evidence informing exercise, function, smoking, sleep, weight, or stress evidence. Routed as observational or registry; applicability was limited to the population and design reported.

097 Sleep patterns, physical activity, genetic susceptibility, and incident rheumatoid arthritis: a prospective cohort study.

PMID 39272142 • 2024 • observational or registry • lifestyle and function

Population/registry evidence informing exercise, function, smoking, sleep, weight, or stress evidence. Routed as observational or registry; applicability was limited to the population and design reported.

098 Features of sarcopenic obesity in rheumatoid arthritis patients.

PMID 40847783 • 2025 • other peer reviewed research • lifestyle and function

Peer-reviewed evidence informing exercise, function, smoking, sleep, weight, or stress evidence. Routed as other peer reviewed research; applicability was limited to the population and design reported.

099 Rheumatoid arthritis treatment: Is exercise a game changer?

PMID 40028397 • 2024 • review • lifestyle and function

Review used to map exercise, function, smoking, sleep, weight, or stress evidence. Routed as review; applicability was limited to the population and design reported.

100 Physical activity in patients with rheumatoid arthritis.

PMID 26639220 • 2016 • review • lifestyle and function

Review used to map exercise, function, smoking, sleep, weight, or stress evidence. Routed as review; applicability was limited to the population and design reported.

Appendix B

Complete 87-paper arXiv review catalogue

The broad arXiv source was exhausted at 87 unique results. Thirty-three papers met the transparent treatment screen; no new compatible public baseline treatment-response cohort was identified.

- 001 MechAInstic: An LLM-guided Multi-Agent System for Reasoning over Genome-Scale Constraint-Based Metabolic Models**
arXiv 2607.18249v1 • 2026-05-08 • candidate ra treatment research
Potential data lead retained for audit; population, exposure, baseline, outcome, or access requirements were incomplete.
- 002 RAM-H1200: A Unified Evaluation and Dataset on Hand Radiographs for Rheumatoid Arthritis**
arXiv 2605.05616v1 • 2026-05-07 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 003 Chronological Contrastive Learning: Few-Shot Progression Assessment in Irreversible Diseases**
arXiv 2603.21935v2 • 2026-03-23 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 004 Image-based Joint-level Detection for Inflammation in Rheumatoid Arthritis from Small and Imbalanced Data**
arXiv 2602.14365v1 • 2026-02-16 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 005 Network-based prediction of drug combinations with quantum annealing**
arXiv 2512.20199v1 • 2025-12-23 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 006 Automated Radiographic Total Sharp Score (ARTSS) in Rheumatoid Arthritis: A Solution to Reduce Inter-Intra Reader Variation and Enhancing Clinical Practice**
arXiv 2509.06854v1 • 2025-09-08 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 007 Network Community Detection and Novelty Scoring Reveal Underexplored Hub Genes in Rheumatoid Arthritis**
arXiv 2509.00897v1 • 2025-08-31 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 008 Interpretable Rheumatoid Arthritis Scoring via Anatomy-aware Multiple Instance Learning**
arXiv 2508.06218v1 • 2025-08-08 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 009 Pragmatic Policy Development via Interpretable Behavior Cloning**
arXiv 2507.17056v1 • 2025-07-22 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 010 Challenging Disability and Interaction Norms in XR: Cooling Down the Empathy Machine in Waiting for Hands**
arXiv 2507.15481v1 • 2025-07-21 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

- 011 Using off-treatment sequential multiple imputation for binary outcomes to address intercurrent events handled by a treatment policy strategy**
arXiv 2507.14006v1 • 2025-07-18 • candidate ra treatment research
Potential data lead retained for audit; population, exposure, baseline, outcome, or access requirements were incomplete.
- 012 RAM-W600: A Multi-Task Wrist Dataset and Benchmark for Rheumatoid Arthritis**
arXiv 2507.05193v4 • 2025-07-07 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 013 Classification of autoimmune diseases from Peripheral blood TCR repertoires by multimodal multi-instance learning**
arXiv 2507.04981v4 • 2025-07-07 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 014 Evaluation of clinical utility in emulated clinical trials**
arXiv 2506.03991v2 • 2025-06-04 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 015 Right Prediction, Wrong Reasoning: Uncovering LLM Misalignment in RA Disease Diagnosis**
arXiv 2504.06581v1 • 2025-04-09 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 016 BioX-CPATH: Biologically-driven Explainable Diagnostics for Multistain IHC Computational Pathology**
arXiv 2503.20880v2 • 2025-03-26 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 017 Layer Separation: Adjustable Joint Space Width Images Synthesis in Conventional Radiography**
arXiv 2502.01972v1 • 2025-02-04 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 018 Effect of a new type of healthy and live food supplement on osteoporosis blood parameters and induced rheumatoid arthritis in Wistar rats**
arXiv 2501.19343v1 • 2025-01-31 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 019 Hengqin-RA-v1: Advanced Large Language Model for Diagnosis and Treatment of Rheumatoid Arthritis with Dataset based Traditional Chinese Medicine**
arXiv 2501.02471v2 • 2025-01-05 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 020 Going Beyond H&E and Oncology: How Do Histopathology Foundation Models Perform for Multi-stain IHC and Immunology?**
arXiv 2410.21560v1 • 2024-10-28 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 021 A Comparative Study of Multiple Deep Learning Algorithms for Efficient Localization of Bone Joints in the Upper Limbs of Human Body**
arXiv 2410.20639v1 • 2024-10-28 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 022 Dynamic borrowing from historical controls via the synthetic prior with covariates in randomized clinical trials**
arXiv 2410.07242v2 • 2024-10-07 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

023 Group lasso based selection for high-dimensional mediation analysis

arXiv 2409.20036v2 • 2024-09-30 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

024 Incremental Causal Effect for Time to Treatment Initialization

arXiv 2409.13097v2 • 2024-09-19 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

025 Federated One-Shot Ensemble Clustering

arXiv 2409.08396v1 • 2024-09-12 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

026 Associations between exposure to OPEs and rheumatoid arthritis risk among adults in NHANES, 2011-2018

arXiv 2409.00745v1 • 2024-09-01 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

027 Deep Learning Models to Automate the Scoring of Hand Radiographs for Rheumatoid Arthritis

arXiv 2406.09980v1 • 2024-06-14 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

028 An adaptive enrichment design using Bayesian model averaging for selection and threshold-identification of tailoring variables

arXiv 2405.08180v1 • 2024-05-13 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

029 How to develop, externally validate, and update multinomial prediction models

arXiv 2312.12008v2 • 2023-12-19 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

030 Automated segmentation of rheumatoid arthritis immunohistochemistry stained synovial tissue

arXiv 2309.07255v1 • 2023-09-13 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

031 Microbiome-derived bile acids contribute to elevated antigenic response and bone erosion in rheumatoid arthritis

arXiv 2307.08848v1 • 2023-07-14 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

032 MIRACLE: Multi-task Learning based Interpretable Regulation of Autoimmune Diseases through Common Latent Epigenetics

arXiv 2306.13866v2 • 2023-06-24 • candidate ra treatment research

Dataset accession(s) were checked; the identified series did not meet the public baseline RA treatment-response contract.

033 A Deep Registration Method for Accurate Quantification of Joint Space Narrowing Progression in Rheumatoid Arthritis

arXiv 2304.13938v1 • 2023-04-27 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

034 Estimation of age-specific excess mortality of men and women with rheumatoid arthritis (RA) in Germany

arXiv 2302.11576v1 • 2023-02-22 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

- 035 multi-GPA-Tree: Statistical Approach for Pleiotropy Informed and Functional Annotation Tree Guided Prioritization of GWAS Results**
arXiv 2302.01982v1 • 2023-02-03 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 036 Nanoparticles Passive Targeting Allows Optical Imaging of Bone Diseases**
arXiv 2301.01896v1 • 2023-01-05 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 037 Prediction of drug effectiveness in rheumatoid arthritis patients based on machine learning algorithms**
arXiv 2210.08016v3 • 2022-10-14 • candidate ra treatment research
Direct response-research lead, but the named registry was not supplied as a reproducible public patient-level dataset.
- 038 Clinical Utility Gains from Incorporating Comorbidity and Geographic Location Information into Risk Estimation Equations for Atherosclerotic Cardiovascular Disease**
arXiv 2209.06985v2 • 2022-09-15 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 039 Semi-supervised Transfer Learning for Evaluation of Model Classification Performance**
arXiv 2208.07927v2 • 2022-08-16 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 040 Employing Feature Selection Algorithms to Determine the Immune State of a Mouse Model of Rheumatoid Arthritis**
arXiv 2207.05882v2 • 2022-07-12 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 041 CNN-based fully automatic wrist cartilage volume quantification in MR Image**
arXiv 2206.11127v1 • 2022-06-22 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 042 A Sub-pixel Accurate Quantification of Joint Space Narrowing Progression in Rheumatoid Arthritis**
arXiv 2205.09315v2 • 2022-05-19 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 043 A robust Bayesian bias-adjusted random effects model for consideration of uncertainty about bias terms in evidence synthesis**
arXiv 2204.10645v1 • 2022-04-22 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 044 Bridging disconnected networks of first and second lines of biologic therapies in rheumatoid arthritis with registry data: Bayesian evidence synthesis with target trial emulation**
arXiv 2201.01720v1 • 2022-01-05 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.
- 045 Towards Super-Resolution CEST MRI for Visualization of Small Structures**
arXiv 2112.01905v1 • 2021-12-03 • out of scope or background only
Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.
- 046 Rheumatoid Arthritis: Automated Scoring of Radiographic Joint Damage**
arXiv 2110.08812v1 • 2021-10-17 • candidate ra treatment research
Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

047 Deep Learning for Rheumatoid Arthritis: Joint Detection and Damage Scoring in X-rays

arXiv 2104.13915v2 • 2021-04-28 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

048 DeepRA: Predicting Joint Damage From Radiographs Using CNN with Attention

arXiv 2102.06982v2 • 2021-02-13 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

049 Topological data analysis distinguishes parameter regimes in the Anderson-Chaplain model of angiogenesis

arXiv 2101.00523v2 • 2021-01-02 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

050 MixTwice: large-scale hypothesis testing for peptide arrays by variance mixing

arXiv 2011.07420v1 • 2020-11-15 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

051 COMO: A Pipeline for Multi-Omics Data Integration in Metabolic Modeling and Drug Discovery

arXiv 2011.02103v2 • 2020-11-04 • candidate ra treatment research

Dataset accession(s) were checked; the identified series did not meet the public baseline RA treatment-response contract.

052 Automatic Chronic Degenerative Diseases Identification Using Enteric Nervous System Images

arXiv 2011.00160v1 • 2020-10-31 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

053 Augmented Transfer Regression Learning with Semi-non-parametric Nuisance Models

arXiv 2010.02521v3 • 2020-10-06 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

054 A Dynamic Deep Neural Network For Multimodal Clinical Data Analysis

arXiv 2008.06294v1 • 2020-08-14 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

055 An one-factor copula mixed model for joint meta-analysis of multiple diagnostic tests

arXiv 2006.09278v2 • 2020-06-16 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

056 Blocking of the CD80/86 axis as a therapeutic approach to prevent progression to more severe forms of COVID-19

arXiv 2005.10055v1 • 2020-05-20 • candidate ra treatment research

Dataset accession(s) were checked; the identified series did not meet the public baseline RA treatment-response contract.

057 Prior Adaptive Semi-supervised Learning with Application to EHR Phenotyping

arXiv 2003.11744v2 • 2020-03-26 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

058 Quantitative predictive modelling approaches to understanding rheumatoid arthritis: A brief review

arXiv 1911.09035v4 • 2019-11-20 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

059 Cartilage-binding antibodies induce pain through immune complex-mediated activation of neurons

arXiv 1908.05298v1 • 2019-08-08 • candidate ra treatment research

Potential data lead retained for audit; population, exposure, baseline, outcome, or access requirements were incomplete.

060 An Evaluation Toolkit to Guide Model Selection and Cohort Definition in Causal Inference

arXiv 1906.00442v1 • 2019-06-02 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

061 Optimal Statistical Inference for Individualized Treatment Effects in High-dimensional Models

arXiv 1904.12891v2 • 2019-04-29 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

062 Hand range of motion evaluation for Rheumatoid Arthritis patients

arXiv 1903.06949v1 • 2019-03-16 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

063 Pain pathogenesis in rheumatoid arthritis -- what have we learned from animal models

arXiv 1903.04987v1 • 2019-03-12 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

064 Empowering individual trait prediction using interactions

arXiv 1901.08814v1 • 2019-01-25 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

065 Interpretable Graph Convolutional Neural Networks for Inference on Noisy Knowledge Graphs

arXiv 1812.00279v1 • 2018-12-01 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

066 JS-MA: A Jensen-Shannon Divergence Based Method for Mapping Genome-wide Associations on Multiple Diseases

arXiv 1811.07099v1 • 2018-11-17 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

067 Unsupervised Ensemble Learning via Ising Model Approximation with Application to Phenotyping Prediction

arXiv 1810.06376v1 • 2018-10-15 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

068 Using routinely collected patient data to support clinical trials research in accountable care organizations

arXiv 1807.00668v1 • 2018-06-25 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

069 Variable domain N-linked glycosylation and negative surface charge are key features of monoclonal ACPA: implications for B-cell selection

arXiv 1802.10401v1 • 2018-02-28 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

070 A Bayesian Joint model for Longitudinal DAS28 Scores and Competing Risk Informative Drop Out in a Rheumatoid Arthritis Clinical Trial

arXiv 1801.08628v1 • 2018-01-25 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

071 Semi-Supervised Approaches to Efficient Evaluation of Model Prediction Performance

arXiv 1711.05663v1 • 2017-11-15 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

072 Autoreactivity to malondialdehyde-modifications in rheumatoid arthritis is linked to disease activity and synovial pathogenesis

arXiv 1710.10861v1 • 2017-10-30 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

073 LPG: a four-groups probabilistic approach to leveraging pleiotropy in genome-wide association studies

arXiv 1710.09551v1 • 2017-10-26 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

074 Good Arm Identification via Bandit Feedback

arXiv 1710.06360v2 • 2017-10-17 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

075 Development of a passive Rehabilitation Robot for the wrist joint through the implementation of an Arduino UNO microcontroller

arXiv 1706.05076v1 • 2017-06-07 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

076 Vulnerability of geriatric patients to biomaterial associated infections: in vitro study of biofilm formation by Pseudomonas aeruginosa on orthopedic implants

arXiv 1511.06969v1 • 2015-11-22 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

077 A method for delineation of bone surfaces in photoacoustic computed tomography of the finger

arXiv 1506.02165v1 • 2015-06-06 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

078 Initial results of finger imaging using Photoacoustic Computed Tomography

arXiv 1406.5500v1 • 2014-06-20 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

079 Implementing Evidential Reasoning in Expert Systems

arXiv 1304.2731v1 • 2013-03-27 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

080 Finding the basic neighborhood in variable range Markov random fields: application in SNP association studies

arXiv 1302.5589v1 • 2013-02-22 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

081 Attribute Exploration of Gene Regulatory Processes

arXiv 1204.1995v1 • 2012-04-09 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

082 Detection of treatment effects by covariate-adjusted expected shortfall

arXiv 1101.1407v1 • 2011-01-07 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

083 BOOST: A fast approach to detecting gene-gene interactions in genome-wide case-control studies

arXiv 1001.5130v1 • 2010-01-28 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

084 Peptide strings clues to the genesis and treatment of rheumatoid arthritis: rebuilding self-protective immunity amid fungal ruins

arXiv 0808.1283v1 • 2008-08-10 • candidate ra treatment research

Treatment-relevant paper read for hypothesis generation; no compatible public response dataset or patient-level treatment proof was established.

085 Across and beyond the cell are peptide strings

arXiv 0711.0202v1 • 2007-11-01 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

086 Effective Sample Size: Quick Estimation of the Effect of Related Samples in Genetic Case-Control Association Analyses

arXiv 0611093v3 • 2006-11-28 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

087 LASSO-Patternsearch algorithm with application to ophthalmology and genomic data

arXiv 0610916v2 • 2006-10-30 • out of scope or background only

Read and excluded from direct treatment-response evidence because RA treatment was background, incidental, preclinical, diagnostic, or otherwise out of scope.

Closing research message: the most useful idea to spread is a better evidence contract—objective inflammation, explicit treatment-failure labels, coordinated RA-IBD outcomes, complete safety variables, and independent validation. That foundation is more likely to improve remission research than another unvalidated small-cohort model or cure claim.

